

CHAPTER 7

SMOOTHNESS

Introduction

In this chapter we fix an o-minimal structure $(R, <, \mathcal{S})$ that expands an ordered field $(R, <, 0, 1, +, -, \cdot)$. Recall from Chapter 1, (4.6) that this ordered field is then necessarily real closed. In Section 1 we discuss differentiability, in Section 2 we show that definable functions in one variable are piecewise differentiable and that definable C^1 -maps in several variables satisfy the implicit function theorem. This allows us to obtain in Section 3 the decomposition of definable sets into “smooth” cells. The smooth cell decomposition is then used in the last section to prove the good directions lemma, an important tool in triangulating definable sets in the next chapter.

§1. Differentiability in ordered fields

(1.1) We start by recording some elementary calculus facts valid over arbitrary ordered fields $(R, <, 0, 1, +, -, \cdot)$. (No o-minimality assumptions or definability assumptions on sets and functions are needed in this section.) To shorten notations we denote this ordered field just by R . We equip R with the interval topology, and each set R^m with the corresponding product topology, and we put $|x| := \sup\{|x_i| : i = 1, \dots, m\}$ for $x = (x_1, \dots, x_m) \in R^m$. We also equip R^m with the usual dot product: $x \cdot y = x_1 y_1 + \dots + x_m y_m$. In this section we omit proofs since they are the same as for the field of real numbers.

(1.2) DEFINITION. Let $I \subseteq R$ be open. A function $f : I \rightarrow R^n$ is said to be **differentiable at a point $x \in I$ with derivative $a \in R^n$** if

$$\lim_{t \rightarrow 0} t^{-1} (f(x+t) - f(x)) = a.$$

Note that then f is continuous at x and that a is unique; we write $a = f'(x)$.

(1.3) Let the functions $f, g : I \rightarrow R^n$ be differentiable at x . Then the sum $f + g$ and the dot product $f \cdot g$ are differentiable at x and

$$\begin{aligned}(f + g)'(x) &= f'(x) + g'(x), \\ (f \cdot g)'(x) &= f'(x) \cdot g(x) + f(x) \cdot g'(x).\end{aligned}$$

If moreover $n = 1$ and g does not vanish on I then f/g is differentiable at x and

$$(f/g)'(x) = (f'(x) \cdot g(x) - f(x) \cdot g'(x))/g^2(x).$$

Constant maps are differentiable everywhere with derivative 0.

The identity function on R is differentiable everywhere with derivative 1.

Let I, J be open subsets of R , let $f: I \rightarrow R$ be continuous and differentiable at the point $x \in I$, $g: J \rightarrow R$ differentiable at $f(x) \in J$. Then $g \circ f$, defined on the open set $I \cap f^{-1}(J)$, is differentiable at x and

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x).$$

(1.4) DIRECTIONAL DERIVATIVES. Consider now a map $f: U \rightarrow R^n$ with $U \subseteq R^m$ open, a point $x \in U$, and a vector $v \in R^m$. We say that f is **differentiable at x in the v -direction with derivative $a \in R^n$** if the function $t \mapsto f(x + tv)$ (defined on an open neighborhood of $0 \in R^m$) is differentiable at 0 with derivative a , that is

$$\lim_{t \rightarrow 0} t^{-1}(f(x + tv) - f(x)) = a.$$

In that case we write $d_x f(v) = a$. Let $e(1), \dots, e(m)$ be the standard basis of the R -linear space R^m , so $e(1) = (1, 0, \dots, 0)$, etc. As in the case of the real field we write $(\partial f / \partial x_i)(x)$ for $d_x f(e(i))$ and call it the i^{th} partial of f at x .

(1.5) THE DIFFERENTIAL OF A MAP. Let $f = (f_1, \dots, f_n): U \rightarrow R^n$ be a map on an open subset U of R^m . Let $x \in U$ and $T: R^m \rightarrow R^n$ be a linear map.

We call f **differentiable at x with differential T** if for each $\epsilon > 0$ we have

$$|f(x + v) - f(x) - T(v)| < \epsilon |v|,$$

for all sufficiently small vectors $v \in R^m$. Then f is continuous at x and T is unique; we write $d_x f = T$. This notation is consistent with that in (1.4), since if T is as above, then f is differentiable at x in the v -direction for each v , and $T(v)$ = directional derivative of f at x in the v -direction. For $m = 1$ this notion of differentiability coincides with the one in (1.2), with $d_x f(1) = f'(x)$.

The map $f = (f_1, \dots, f_n)$ is differentiable at x iff each coordinate function $f_i: U \rightarrow R$ is differentiable at x . In that case all partials $(\partial f_i / \partial x_j)(x)$ exist and the $n \times m$ matrix $(\partial f_i / \partial x_j)(x)$ is exactly the matrix of $d_x f$ relative to the standard bases. It is called the **Jacobian matrix of f at x** .

(1.6) BEHAVIOR UNDER ALGEBRAIC OPERATIONS. If the maps $f, g: U \rightarrow R^n$ are differentiable at $x \in U$, U open in R^m , then $f + g$ and cf , for each $c \in R$, are differentiable at x and

$$\begin{aligned} d_x(f + g) &= d_x f + d_x g, \\ d_x cf &= c \cdot d_x f. \end{aligned}$$

If also $h: V \rightarrow R^p$ is differentiable at $f(x) \in V$, V open in R^n and f continuous, then $h \circ f$, defined on $U \cap f^{-1}V$, is differentiable at x and

$$d_x(h \circ f) = (d_{f(x)}h) \circ d_x f.$$

Each R -linear map $R^m \rightarrow R^n$ is differentiable at each point in R^m with itself as differential.

§2. Inverse function theorem

(2.1) CONVENTION FOR THE REST OF THIS CHAPTER.

$\mathcal{R} = (R, <, \mathfrak{S})$ is an o-minimal expansion of an ordered field $(R, <, 0, 1, +, -, \cdot)$. (This ordered field is then in fact real closed by Chapter 1, (4.6).) We let $\|x\| := (x \cdot x)^{1/2}$ denote the usual ‘euclidean norm’ of the vector $x \in R^m$.

(2.2) LEMMA (ROLLE). *Suppose $a < b$ in R and the function $f: [a, b] \rightarrow R$ is definable, continuous, $f(a) = f(b)$, and f is differentiable at each point of (a, b) . Then $f'(c) = 0$ for some $c \in (a, b)$.*

PROOF. Take c , $a < c < b$, such that $f(c)$ is maximal or minimal. Then clearly $f'(c) = 0$. $\square$

This has the following easy consequences.

(2.3) MEAN VALUE THEOREM. *Suppose $a < b$ in R , $f: [a, b] \rightarrow R$ is definable and continuous, and differentiable at each point of (a, b) . Then for some $c \in (a, b)$, we have $f(b) - f(a) = (b - a) \cdot f'(c)$.*

(2.4) THEOREM ON CONSTANTS. *Under the same assumptions as in the mean value theorem, suppose that moreover $f'(x) = 0$ for all $x \in (a, b)$. Then f is constant.*

In the following I denotes an interval. Our first goal is to prove

(2.5) PROPOSITION. *If $f: I \rightarrow R$ is definable, then f is differentiable at all but finitely many points of I .*

This requires several lemmas.

LEMMA 1. *Let $f: I \rightarrow R$ be definable. Then for each $x \in I$ the limits*

$$\begin{aligned} f'(x^+) &:= \lim_{t \downarrow 0} t^{-1} (f(x+t) - f(x)), \\ f'(x^-) &:= \lim_{t \uparrow 0} t^{-1} (f(x+t) - f(x)) \end{aligned}$$

exist in R_∞ . If moreover f is continuous and $f'(x^+) > 0$ for all x , then f is strictly increasing and its inverse $f^{-1}: f(I) \rightarrow R$ satisfies $(f^{-1})'(y^+) = 1/f'(x^+)$, for $x \in I$ and $f(x) = y \in f(I)$. (Here $1/+\infty := 0$.)

PROOF. For fixed x the function $g: t \mapsto (f(x+t) - f(x))/t$, defined on an interval $(0, \epsilon)$, is definable, whence the limit $f'(x^+) = \lim_{t \rightarrow 0} g(t)$ exists by Chapter 3, (1.6). Similarly for $f'(x^-)$.

Suppose now that $f'(x^+) > 0$ for all x , and f is continuous. If f were not strictly increasing then f would be constant or strictly decreasing on some subinterval, contradicting $f'(x^+) > 0$ on that subinterval. The formula for the right derivative of f^{-1} is obtained in the usual way.

LEMMA 2. Let $f: I \rightarrow R$ be definable and continuous, and suppose the maps $x \mapsto f'(x^+)$ and $x \mapsto f'(x^-)$ are R -valued and continuous on I . Then f is differentiable at each point of I , and $f': I \rightarrow R$ is continuous.

PROOF. It suffices to show that $f'(a^+) = f'(a^-)$ for all $a \in I$. Suppose the contrary, say $f'(a^+) > f'(a^-)$ for a certain $a \in I$. Then there are $c \in R$ and a subinterval J of I around a such that $f'(x^+) > c > f'(x^-)$ on J . Hence the definable function $g: J \rightarrow R$ given by $g(x) := f(x) - cx$ has the property that $g'(x^+) > 0$, $g'(x^-) < 0$ for all x , so g would be both strictly increasing and strictly decreasing on J . Contradiction.

LEMMA 3. Let $f: I \rightarrow R$ be definable. Then there are only finitely many $x \in I$ such that $f'(x) \in \{-\infty, +\infty\}$.

PROOF. Suppose the definable set $\{x \in I : f'(x^+) = +\infty\}$ is infinite. Then this set contains a whole interval, and for the sake of deriving a contradiction we may as well assume that $f'(x^+) = +\infty$ for all $x \in I$ and that f is continuous. By lemma 1 this implies that f is strictly increasing, hence $f'(x^-) \geq 0$ for all $x \in I$.

After further shrinking the interval we may also assume that we are in one of the following cases:

- (i) $f'(x^-) = +\infty$ for all x in I ,
- (ii) $f'(x^-) \in R$ for all x in I , and the map $x \mapsto f'(x^-)$ is continuous on I .

In case (i) the inverse of f satisfies $(f^{-1})'(y^-) = (f^{-1})'(y^+) = 0$ for all $y \in f(I)$, so that by the theorem on constants f^{-1} is constant, contradicting the injectivity of f^{-1} . In case (ii) we can apply the same argument as in the proof of lemma 2 to get a contradiction.

PROOF OF (2.5). By the monotonicity theorem and lemma 3 we can reduce to the case that f is continuous, and $f'(x^+)$ and $f'(x^-)$ are R -valued and continuous functions of $x \in I$. Now apply lemma 2. $\square$

(2.6) CONTINUOUS DIFFERENTIABILITY.

In the following definition and lemmas we denote by $f = (f_1, \dots, f_n): U \rightarrow R^n$ a definable map on a (definable) open set $U \subseteq R^m$. We also put

$$\text{Lin}(R^m, R^n) := R\text{-linear space of } R\text{-linear maps } R^m \rightarrow R^n.$$

DEFINITION. We call f a C^1 -**map** if the partials $(\partial f_i / \partial x_j)$ are defined as R -valued functions on U and are continuous. (It follows that f is continuous.)

LEMMA. If f is C^1 , then f is differentiable at each point of U , and the map

$$x \mapsto d_x f: U \rightarrow \text{Lin}(R^m, R^n) \cong R^{n \times m}$$

is continuous. Conversely, if f is differentiable at each point of U and the map $x \mapsto d_x f: U \rightarrow R^{n \times m}$ is continuous, then f is a C^1 -map.

(Here we identify each R -linear map from R^m to R^n with its $n \times m$ matrix relative to the standard bases.)

The second part of the lemma is an immediate consequence of the fact stated at the end of (1.5). The proof of the first part follows the classical proof, which depends on the mean value theorem. (That is why we restrict ourselves to definable maps, for which this result is available.)

Next we define the norm of a linear map.

(2.7) DEFINITION. For an R -linear map $T: R^m \rightarrow R^n$ we put

$$|T| := \max\{|Tx| : |x| \leq 1, x \in R^m\}.$$

(Note that the maximum exists since T is definable and continuous and that $|T(x)| \leq |T| \cdot |x|$ for all $x \in R^m$.)

(2.8) LEMMA. Let $f: U \rightarrow R^n$ be C^1 , and let $[a, b] := \{(1-t)a + tb : 0 \leq t \leq 1\}$ be a line segment contained in U . Then

$$|f(b) - f(a)| \leq |b - a| \cdot \max_{y \in [a, b]} |d_y f|.$$

PROOF. Define $g: [0, 1] \rightarrow R^n$ by $g(t) := f((1-t)a + tb)$. Then g is differentiable at each t , $0 < t < 1$, with $g'(t) = d_y f(b - a)$, where $y := (1-t)a + tb$, so $|g'(t)| \leq M$, with

$$M := |b - a| \cdot \max_{y \in [a, b]} |d_y f|.$$

Then clearly by the mean value theorem, $|f(b) - f(a)| = |g(1) - g(0)| \leq M$. $\square$

(2.9) LEMMA. *With the same assumptions as in the previous lemma, let $x \in U$. Then*

$$|f(b) - f(a) - d_x f(b - a)| \leq |b - a| \cdot \max_{y \in [a, b]} |d_y f - d_x f|.$$

PROOF. Apply the previous lemma to the map $y \mapsto f(y) - d_x f(y)$. $\square$

(2.10) LEMMA. *With the same assumptions, let $m = n$, $a \in U$ and suppose $d_a f$ is invertible. Then there are $\epsilon > 0$ and $c > 0$ in R such that*

$$|f(x) - f(y)| \geq c|x - y| \text{ for all } x, y \in U \text{ with } |x - a|, |y - a| < \epsilon.$$

In particular f is injective on a neighborhood of a .

PROOF. Take $\epsilon > 0$ so small that the open ball $B(a, \epsilon)$ is contained in U . Let $x, y \in B(a, \epsilon)$. Applying the previous lemma, we have

$$|f(x) - f(y) - d_a f(x - y)| \leq |x - y| \cdot \max_{z \in [x, y]} |d_z f - d_a f|,$$

so

$$|f(x) - f(y)| \geq |d_a f(x - y)| - |x - y| \cdot \max_{z \in [x, y]} |d_z f - d_a f|.$$

Since $d_a f$ is invertible there is $c' > 0$, independent of x, y , such that

$$|d_a f(x - y)| \geq c'|x - y|.$$

(Use $|z| \geq c'|T(z)|$ for $T = (d_a f)^{-1}$, $z = d_a f(x - y)$.) Decreasing ϵ if necessary we may also assume that

$$|d_b f - d_a f| < c'/2 \text{ for all } b \in B(a, \epsilon).$$

Hence

$$\begin{aligned} |f(x) - f(y)| &\geq c'|x - y| - (c'/2)|x - y| \\ &\geq c|x - y| \end{aligned}$$

for $c = c'/2$. $\square$

(2.11) INVERSE FUNCTION THEOREM. *Let $f: U \rightarrow R^m$ be a definable C^1 -map on a definable open set $U \subseteq R^m$ and $a \in U$ a point where $d_a f: R^m \rightarrow R^m$ is invertible. Then there are definable open neighborhoods $U' \subseteq U$ of a and V' of $f(a)$ such that f maps U' homeomorphically onto V' and $f^{-1}: V' \rightarrow U'$ is also C^1 .*

PROOF. In the statement of lemma (2.10) we may of course replace the norm $|\cdot|$ by the equivalent euclidean norm $\|\cdot\|$, which is more convenient here. Take $\epsilon, c > 0$ such that

$$\begin{aligned} \|x - a\| \leq \epsilon &\Rightarrow x \in U \text{ and } d_x f \text{ is invertible,} \\ \text{and } \|x - a\|, \|y - a\| \leq \epsilon &\Rightarrow \|f(x) - f(y)\| \geq c\|x - y\|. \end{aligned}$$

In particular, $\|x - a\| = \epsilon \Rightarrow \|f(x) - f(a)\| \geq c\epsilon$. We claim that

$$\{y : \|y - f(a)\| < (1/2)c\epsilon\} \subseteq \{f(x) : \|x - a\| < \epsilon\}.$$

To see this, let $\|y - f(a)\| < (1/2)c\epsilon$ and consider the function

$$P(x) := \|f(x) - y\|^2 = \sum (f_i(x) - y_i)^2 \text{ on the ball } \|x - a\| \leq \epsilon.$$

By Chapter 6, (1.11) the function P assumes a minimum value, but if $\|x - a\| = \epsilon$ then

$$\begin{aligned} P(x) &= \|(f(x) - f(a)) - (y - f(a))\|^2 > (c\epsilon - (1/2)c\epsilon)^2 \\ &= (1/4)c^2\epsilon^2 > \|y - f(a)\|^2 = P(a), \end{aligned}$$

so P must assume its minimum value at an interior point b , $\|b - a\| < \epsilon$. Then

$$0 = (\partial P / \partial x_j)(b) = \sum 2(f_i(b) - y_i) \cdot (\partial f_i / \partial x_j)(b) \text{ for all } j,$$

that is, $d_b f(f(b) - y) = 0$. Since $d_b f$ is invertible this gives $f(b) = y$, and we have proved the claim. Since ϵ can be taken arbitrarily small this argument shows that f maps each neighborhood of a onto a neighborhood of $f(a)$. Put $U' := \{x : \|x - a\| < \epsilon\}$. For the same reason f maps each neighborhood of each point $x \in U'$ onto a neighborhood of $f(x)$. Since f is also injective on U' it follows that f maps U' homeomorphically onto the open set $f(U')$. It remains to show that $f^{-1} : f(U') \rightarrow U'$ is a C^1 -map.

Let $y \in f(U')$ approach $f(a)$ and put $f^{-1}(y) := b \in U'$, so b approaches a . From

$$(f(b) - f(a) - d_a f(b - a)) / \|b - a\| \rightarrow 0 \text{ as } b \rightarrow a,$$

and $\|b - a\| \leq c^{-1}\|y - f(a)\|$ we obtain by applying $(d_a f)^{-1}$

$$((d_a f)^{-1}(y - f(a)) - (f^{-1}(y) - a)) / \|y - f(a)\| \rightarrow 0 \text{ as } y \rightarrow f(a).$$

Hence f^{-1} is differentiable at $f(a)$ with $d_{f(a)} f^{-1} = (d_a f)^{-1}$. For the same reasons f^{-1} is differentiable at every point $y \in f(U')$ with $d_y f^{-1} = (d_{f^{-1}(y)} f)^{-1}$. Thus f^{-1} is C^1 . $\square$

COROLLARY (IMPLICIT FUNCTION THEOREM). *Let $U \subseteq \mathbb{R}^{m+n}$ be a definable open set and $f_1, \dots, f_n : U \rightarrow \mathbb{R}$ definable C^1 -functions. Let $(x_0, y_0) \in U$ be such that $f_1(x_0, y_0) = \dots = f_n(x_0, y_0) = 0$ and the $n \times n$ matrix $((\partial f_j / \partial y_k)(x_0, y_0))_{1 \leq j, k \leq n}$ is invertible. Then there are open definable neighborhoods V of x_0 in $\mathbb{R}^m$ and W of y_0 in $\mathbb{R}^n$ respectively, and there is a definable C^1 -map $\phi : V \rightarrow W$ such that $V \times W \subseteq U$, $\phi(x_0) = y_0$, and such that for all $(x, y) \in V \times W$ we have*

$$f_1(x, y) = \dots = f_n(x, y) = 0 \Leftrightarrow y = \phi(x).$$

PROOF. Apply the inverse function theorem to the map

$$(x, y) \mapsto (x, f_1(x, y), \dots, f_n(x, y)): U \rightarrow R^{m+n}. \quad \square$$

(2.12) EXERCISES.

1. (L'Hôpital's rule) Let I be an interval and $f, g: I \rightarrow R$ definable functions, and let a be one of the endpoints of the interval, possibly $a = +\infty$ or $a = -\infty$. Suppose that $g'(x) \neq 0$ for all $x \in I$ in some neighborhood of a , and that $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$, or $\lim_{x \rightarrow a} |f(x)| = \lim_{x \rightarrow a} |g(x)| = +\infty$. Then

$$\lim_{x \rightarrow a} (f(x)/g(x)) = \lim_{x \rightarrow a} (f'(x)/g'(x)).$$

(Note that both limits exist in R_∞ , by Chapter 3, (1.6).)

2. (Taylor's formula) Suppose the definable function $f: I \rightarrow R$ is $(n+1)$ times differentiable on the interval I , and let $a, b \in I$, $a < b$. Then

$$\begin{aligned} f(b) = f(a) + f'(a)(b-a) + \frac{f^{(2)}(a)}{2!}(b-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(b-a)^n \\ + \frac{f^{(n+1)}(z)}{(n+1)!}(b-a)^{n+1} \end{aligned}$$

for some z with $a < z < b$.

§3. Definable maps are piecewise C^1

(3.1) Another basic tool is a C^1 -version of cell decomposition. First we extend the notion of C^1 -map and define C^1 -cells.

DEFINITION.

- (1) A definable map $f: A \rightarrow R^n$, where $A \subseteq R^m$ is not necessarily open, is said to be a **C^1 -map** if there are a definable open set $U \subseteq R^m$ containing A and a definable C^1 -map $F: U \rightarrow R^n$ such that $F|_A = f$. (Then f is continuous, and for open A this gives the usual notion of C^1 -map on an open set.)
- (2) The notion of **C^1 -cell** is defined inductively as in Chapter 3, (2.3), except that when forming $\Gamma(f)$ and (f, g) we now require the functions f and g (when R -valued) to be C^1 (and definable) instead of just continuous (and definable).

Note that every inclusion map $A \rightarrow R^m$ (for definable $A \subseteq R^m$) is C^1 , and that if $f: A \rightarrow R^n$ is C^1 and $g: B \rightarrow R^p$ is C^1 , with $B \subseteq R^n$, then $g \circ f: f^{-1}(B) \rightarrow R^p$ is C^1 . Also $f = (f_1, \dots, f_n): A \rightarrow R^n$ is C^1 iff each function $f_i: A \rightarrow R$ is C^1 .

(3.2) THEOREM (C^1 -CELL DECOMPOSITION).

- (I_m) For any definable sets $A_1, \dots, A_k \subseteq R^m$ there is a decomposition of R^m into C^1 -cells partitioning $A_1, \dots, A_k$.
- (II_m) For every definable function $f: A \rightarrow R$, $A \subseteq R^m$, there is a decomposition of R^m into C^1 -cells, partitioning A , such that each restriction $f|_C: C \rightarrow R$ is C^1 for each cell $C \subseteq A$ of the decomposition.

If f and A are as in (II_m) and p is an interior point of A , we define

$$\nabla f(p) := ((\partial f / \partial x_1)(p), \dots, (\partial f / \partial x_m)(p)),$$

provided these partials exist at p . If some partial is not defined at p , then ∇f is not defined at p . Further we put

$$A' := \{p \in A : p \text{ is an interior point of } A \text{ at which } \nabla f \text{ is defined}\}.$$

Along with (I_m) and (II_m) we will prove the following technical result.

(III_m) $A - A'$ has empty interior.

PROOF OF THE THEOREM. By induction on m : (I₁) is trivial and (III₁) is a reformulation of (2.5). To prove (II₁), let $f: A \rightarrow R$, with $A \subseteq R$, be definable. By (2.5) there is a decomposition $\mathcal{D}$ of R partitioning A such that the restriction of f to each interval in $\mathcal{D}$ contained in A is differentiable, and by the monotonicity theorem we may assume, after refining $\mathcal{D}$ if necessary, that the restriction of f to each interval in $\mathcal{D}$ has continuous derivative.

Now we shall assume inductively that (I_d), (II_d) and (III_d) hold for all $d \leq m$, and derive successively (I_{m+1}), (III_{m+1}) and (II_{m+1}).

PROOF OF (I_{m+1}). Let $A_1, \dots, A_k \subseteq R^{m+1}$ be definable. We want to find a C^1 -decomposition of R^{m+1} partitioning $A_1, \dots, A_k$. By ordinary cell decomposition there is a decomposition $\mathcal{D}$ of R^{m+1} partitioning $A_1, \dots, A_k$. Then $\pi(\mathcal{D})$ is a decomposition of R^m where $\pi: R^{m+1} \rightarrow R^m$ is the usual projection map. Let $\pi(\mathcal{D}) = \{C_1, \dots, C_n\}$, and for each $i = 1, \dots, n$ let the cells of $\mathcal{D}$ that project onto C_i be

$$(-\infty, f_{i1}), \Gamma(f_{i1}), (f_{i1}, f_{i2}), \dots, \Gamma(f_{is}), (f_{is}, +\infty),$$

where $f_{i1}, \dots, f_{is}: C_i \rightarrow R$ are definable and continuous, $s = s(i)$. By (I_m) and (II_m) we may assume, after suitably refining $\pi(\mathcal{D})$, and $\mathcal{D}$ accordingly, that all C_i and all f_{ij} are C^1 . Then $\mathcal{D}$ is a C^1 -decomposition as required. $\square$

PROOF OF (III_{m+1}). Let $f: A \rightarrow R$ be definable, $A \subseteq R^{m+1}$, and define A' as above. Consider an open box $U \times (a, b) \subseteq A$. It suffices to show that $U \times (a, b)$ intersects A' . By (2.5) and definable choice we can pick for each $p \in U$ an interval

$(\alpha(p), \beta(p)) \subseteq (a, b)$ such that α and β are definable R -valued functions on U and $\partial f / \partial x_{m+1}$ is defined on $\{p\} \times (\alpha(p), \beta(p))$. Using cell decomposition we can shrink U so that α and β are continuous on U ; shrinking U further and changing a and b we may as well assume that $\partial f / \partial x_{m+1}$ is defined on the entire box $U \times (a, b)$. Take any $t \in (a, b)$. By applying (III _{m}) to the function $p \mapsto f(p, t): U \rightarrow R$ we see that there must exist $p_0 \in U$ such that all partials $(\partial f / \partial x_i)(p_0, t)$ for $i = 1, \dots, m$ are defined. Hence $(p_0, t) \in A'$, as desired. $\square$

PROOF OF (II _{$m+1$}). Let $f: A \rightarrow R$ be definable, $A \subseteq R^{m+1}$. Let A' be as above. Take a decomposition $\mathcal{D}$ of R^{m+1} partitioning A and A' such that ∇f (and hence f) is continuous on each open cell of $\mathcal{D}$ contained in A' .

CLAIM. For each cell $C \in \mathcal{D}$ with $C \subseteq A$, $\dim(C) < m+1$, there is a decomposition $\mathcal{D}_C$ of R^{m+1} partitioning C such that $f|_D$ is C^1 for each $D \in \mathcal{D}_C$.

PROOF OF CLAIM. Let $C \in \mathcal{D}$, $C \subseteq A$, $\dim(C) = d \leq m$, and consider the homeomorphism $p_C: C \rightarrow p(C)$ onto the open cell $p(C) \subseteq R^d$. By (II _{d}) we can partition $p(C)$ into finitely many cells B such that $f \circ p_C^{-1}|_B$ is C^1 .

By composing with the C^1 -map p_C we obtain that $f|_{p_C^{-1}(B)}$ is C^1 . Take for $\mathcal{D}_C$ a decomposition that partitions each of the sets $p_C^{-1}(B)$. This proves the claim. $\square$

Now we are done: by (I _{$m+1$}) there is a decomposition $\mathcal{D}'$ that refines $\mathcal{D}$ and all $\mathcal{D}_C$ as in the claim, and that consists moreover entirely of C^1 -cells. Let $C' \in \mathcal{D}'$ and $C' \subseteq A$. It suffices to show that $f|_{C'}$ is C^1 . Take the cell $C \in \mathcal{D}$ that contains C' , so $C \subseteq A$. If C is open, then by (III _{$m+1$}) the cell C intersects A' , hence $C \subseteq A'$, so $f|_C$ is C^1 , so $f|_{C'}$ is C^1 . If $C \in \mathcal{D}$ is not open, apply the claim to conclude that $f|_{C'}$ is C^1 . $\square$

(3.3) EXERCISES.

Let $f = (f_1, \dots, f_n): U \rightarrow R^n$ be a definable map on a (definable) open set $U \subseteq R^m$. We define inductively what it means for f to be a C^k -map, where k is a positive integer. For $k = 1$ this has been defined in (2.6). For $k > 1$ the map f is said to be a C^k -map if f is a C^1 -map and $df: U \rightarrow R^{nm}$ is a C^{k-1} -map.

Next, let $f: A \rightarrow R^n$ be a definable map where $A \subseteq R^m$ is not necessarily open. Then we define f to be a C^k -map by replacing in (3.1) everywhere “ C^1 ” by “ C^k ”. Similarly, define C^k -cells as in (3.1) by replacing everywhere “ C^1 ” by “ C^k ”.

1. Show that the remarks at the end of (3.1) go through with “ C^1 ” replaced by “ C^k ”.

2. State and prove the C^k -Cell Decomposition Theorem, $k \geq 1$.

§4. Existence of good directions

(4.1) For each $x \in R^m$ with $\|x\| < 1$, put

$$v(x) := (x_1, \dots, x_m, \sqrt{1 - \|x\|^2}),$$

so that $\|v(x)\| = 1$, $v(x) \in R^{m+1}$. (Note: $v(x)$ is the point on the unit sphere $S^m \subseteq R^{m+1}$ lying directly above x .) Let $A \subseteq R^{m+1}$ be definable with $\dim(A) < m + 1$. We now prove the final result of this chapter, the existence of “good directions” for A . It will be crucial in triangulating definable sets in the next chapter.

(4.2) THEOREM (GOOD DIRECTIONS LEMMA). *Let $B \subseteq R^m$ be a box contained in the disc $\|x\| < 1$. Then there is $x \in B$ such that for each point $p \in R^{m+1}$ the set $\{t \in R : p + t \cdot v(x) \in A\}$ is finite.*

For such x , every line in the affine space R^{m+1} with direction $v(x)$ intersects A in only finitely many points. We then call $v(x)$ a **good** direction. So the theorem tells us that the set of good directions is dense in the unit sphere $S^m \subseteq R^{m+1}$.

PROOF. Suppose there is no such x . Then for each $x \in B$ there is $p \in A$ and $\epsilon > 0$ such that $p + t \cdot v(x) \in A$ for all t with $|t| < \epsilon$. By definable choice there is then a definable map assigning to each $x \in B$ a pair (p, ϵ) with these properties. Applying (3.2) we may even assume, after shrinking B , that there are a C^1 -map $P: B \rightarrow R^{m+1}$ and a fixed $\epsilon > 0$ such that $P(x) + t \cdot v(x) \in A$, for all $x \in B$ and $|t| < \epsilon$. Define the C^1 -map $\phi: B \times R \rightarrow R^{m+1}$ by: $\phi(x, t) := P(x) + t \cdot v(x)$. Let us fix a point $x \in B$.

CLAIM. There is t , $|t| < \epsilon$, such that the R -linear map $d_{(x,t)}\phi: R^{m+1} \rightarrow R^{m+1}$ is invertible.

To prove the claim we consider the matrix $M(t)$ of $d_{(x,t)}\phi$ with respect to the standard basis $e_1, \dots, e_m, e_{m+1}$. (In this context the elements of R^{m+1} are column vectors written as $(a_1, \dots, a_{m+1})^t$, the “t” standing for “transpose”.) Let $P = (P_1, \dots, P_{m+1})$ and put $p_i := ((\partial P_1 / \partial x_i)(x), \dots, (\partial P_{m+1} / \partial x_i)(x))^t$ for $1 \leq i \leq m$. A straightforward computation shows that the columns of $M(t)$ are given by

$$\begin{aligned} M(t) \cdot e_i &= p_i + t \cdot \left(e_i - \left(x_i / \sqrt{1 - \|x\|^2} \right) \cdot e_{m+1} \right), \quad 1 \leq i \leq m, \\ M(t) \cdot e_{m+1} &= v(x). \end{aligned}$$

It follows that $\det(M(t))$ is a polynomial in t . For $t \neq 0$ the $m + 1$ columns of $M(t)$ are independent if and only if the $m + 1$ vectors

$$(p_i/t) + e_i - \left(x_i / \sqrt{1 - \|x\|^2} \right) \cdot e_{m+1} \quad (i = 1, \dots, m), \text{ and } v(x)$$

are independent. When t is large these $m+1$ vectors are close to the $m+1$ vectors

$$e_i - \left(x_i / \sqrt{1 - \|x\|^2} \right) \cdot e_{m+1} \quad (i = 1, \dots, m), \text{ and } v(x),$$

which are easily seen to be independent. So for sufficiently large t the $m+1$ columns of $M(t)$ are independent. Therefore the polynomial $\det(M(t))$ is not identically zero. So $\det(M(t)) \neq 0$ for some t , $|t| < \epsilon$, which proves the claim.

Take such a t , $|t| < \epsilon$, with $d_{(x,t)}\phi$ invertible. From (2.11) it follows that ϕ is injective on a box B_0 around (x, t) . Take B_0 so small that $B_0 \subseteq B \times (-\epsilon, \epsilon)$. Then $\phi(B_0) \subseteq A$ by definition of ϕ . But $\dim(\phi(B_0)) = \dim(B_0) = m+1$, contradicting $\dim(A) < m+1$. $\square$

(4.3) EXERCISES.

1. Let $A \subseteq R^{m+1}$ be a definable set of dimension $\leq m$. Call a unit vector $u \in S^m \subseteq R^{m+1}$ an **asymptotic direction** for A if for each $\epsilon > 0$ and $r > 0$, there is a point $x \in A$ with $\|x\| > r$ and $\|(x/\|x\|) - u\| < \epsilon$. Show that the (definable) set of asymptotic directions for A is of dimension $< m$. (In particular, not every unit vector is an asymptotic direction for A .)

2. Let $A \subseteq R^m$ be definable and $\dim(A) \leq k < m$. Show that there is an $(m-k)$ -dimensional linear subspace L of R^m all of whose translates $v + L$ ($v \in R^m$) meet A in only finitely many points. (Hint: proceed by induction on $m-k$.)

3. In this exercise, assume that R is an ordered field, not necessarily real closed.

Let $A \subseteq R^m$ be semilinear with $\dim(A) < m$. Show that A is contained in a finite union of affine subspaces of R^m of dimension $< m$, and derive that there is a good direction for A , that is, a nonzero vector $u \in R^m$ such that each line $p + Ru$ ($p \in R^m$) intersects A in only finitely many points.

Notes and comments

Proposition (2.5) that definable functions are piecewise differentiable is adapted from the appendix in [19], which also shows C^1 -cell decomposition in the classical real case assuming “strong o-minimality”. The proof of the inverse function theorem is along the lines of that for polynomial maps in Brumfiel [7]. The good directions lemma (4.2) was inspired by a lemma due to Koopman and Brown [36], which also appeared in (unpublished?) lecture notes by Sussmann on subanalytic sets (mid 1980s).